

Holstein–Primakoff magnons coupled to SU(3) flavor-wave bosons on the Pbnm four-sublattice cell of TmFeO₃: linearized equations of motion

Companion to `tmfeo3_notes.tex`
(Dated: May 27, 2026)

We bosonize the mixed Fe ($S=5/2$, SU(2)) plus Tm³⁺ non-Kramers (qutrit, SU(3)) Hamiltonian of TmFeO₃ at finite wavevector. The Fe sector is treated with Holstein–Primakoff (HP) bosons [1] in local Bertaut frames so that the $\Gamma_2(F_x, G_z)$ canted ground state appears as uniform $+\hat{z}_\mu^1$ polarization on every sublattice, with the Klein-four sign structure $\eta_\mu = (+++, +--, -+-, ---)$ absorbed into the local rotations. The Tm CEF qutrit is treated with the generalized SU(3) flavor-wave / multiboson HP construction of Papanicolaou [2], Joshi–Ma–Lhuillier–Mila [3], and Muniz–Kato–Batista [4] (see also the review by Penc and Läuchli [5]): the ground singlet $|E_1\rangle$ is condensed and two flavor magnons $b_\mu^{(2)}, b_\mu^{(3)}$ describe the two remaining transitions. The Fe–Tm bilinear χ exchange and the trilinear W coupling become, after expansion around the classical Γ_2 vacuum, (i) bilinear renormalizations of the Fe magnon and Tm flavor-boson dispersions and (ii) bilinear hybridizations whose sign pattern is dictated by the A_1^+ vs. A_2^+ Bertaut classes of the Gell–Mann operators. After Fourier transformation on the four-sublattice unit cell the linearized equation of motion takes the paraunitary Bogoliubov–de Gennes form $i\partial_t \Psi_{\mathbf{k}} = \tau_3 H(\mathbf{k}) \Psi_{\mathbf{k}}$ with a 24×24 kernel (4 Fe sublattices $\times$ 2 Nambu + 4 Tm sublattices $\times$ 2 flavors $\times$ 2 Nambu). We give all matrix elements explicitly in terms of the microscopic couplings of the implementation in `src/core/unitcell_builders.cpp`.

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I. SETUP AND NOTATION

A. Hamiltonian recap

We start from the microscopic Hamiltonian implemented in `apply_tmfeo3_fe_sector`, `apply_tmfeo3_tm_sector`, and `apply_tmfeo3_fe_tm_couplings` (see also Sec. II of

the companion notes `tmfeo3_notes.tex` [12]):

$$H = H_{\text{FeFe}} + H_{\text{Fe}}^{\text{SIA}} + H_{\text{Fe}}^Z + H_{\text{Tm}}^{\text{CEF}} + H_{\text{Tm}}^Z + H_{\text{TmTm}} + H_{\chi} + H_W. \quad (1)$$

The indices $\mu \in \{0, 1, 2, 3\}$ label the four Fe and the four Tm sublattices in the Pbnm unit cell (Wyckoff 4b and 4c respectively). The Pbnm Klein-four sign matrix is

$$\eta = \begin{pmatrix} + & + & + \\ + & - & - \\ - & + & - \\ - & - & + \end{pmatrix}, \quad \eta_{\mu}^a \in \{\pm 1\}, \quad (2)$$

i.e. the rows give the local-frame signs $(\eta_{\mu}^x, \eta_{\mu}^y, \eta_{\mu}^z)$ for the four sublattices. The Fe local frame is the diagonal matrix

$$D_{\mu} = \text{diag}(\eta_{\mu}^x, \eta_{\mu}^y, \eta_{\mu}^z), \quad \mathbf{S}_{\mu}^{\text{lab}} = D_{\mu} \mathbf{S}_{\mu}^{\text{loc}}. \quad (3)$$

For Tm the SU(3) frame F_{μ} is the 8×8 block $F_{\mu} = \mu_{\text{act}}^{-1} D_{\mu} \mu_{\text{act}}$ on the active+coherence indices and the identity on the population channels; see `tmfeo3_induced_su3_frame` in the source. The Tm 3×3 dipole matrix $\mu_{\text{act}} = (\langle x \rangle, \langle y \rangle, \langle z \rangle)$ on $\{\lambda_2, \lambda_5, \lambda_7\}$ uses the T -symmetric gauge of Sec. III B of [12].

B. Ground state and local rotations

In the Γ_2 phase the Fe spins order in the (F_x, G_z) canted antiferromagnet pattern. In each Fe local frame (3) the staggered G_z component plus the small canted F_x component conspire so that

$$\langle S_{\mu}^{z'} \rangle = +S, \quad \langle S_{\mu}^{x', y'} \rangle = 0, \quad \forall \mu. \quad (4)$$

This is the entire point of the Bertaut local frame: the staggering and the canting angle $\beta_c = \frac{1}{2} \arctan(2D/J)$ are absorbed into the rotation $R_{\mu} = D_{\mu} R_{\beta_c}$ from lab to local frame, leaving a uniform north-pole vacuum suitable for HP expansion (cf. [7, 8]).

For the Tm quirk the Γ_2 ground state is the single occupied CEF level $|E_1\rangle$ at every site. After rotation by F_{μ} this remains $|E_1\rangle$ (the rotation acts trivially on the population channels λ_3, λ_8 , and these alone determine the diagonal projector $|E_1\rangle\langle E_1|$):

$$\langle \lambda_3 \rangle = 1, \quad \langle \lambda_8 \rangle = \frac{1}{\sqrt{3}}, \quad \text{all other } \langle \lambda_a \rangle = 0. \quad (5)$$

II. FE SECTOR: HOLSTEIN-PRIMAKOFF IN LOCAL FRAMES

A. HP expansion

On each Fe site $i \in \mu$ introduce HP bosons $a_{i,\mu}$ with $[a, a^{\dagger}] = 1$ [1]:

$$\begin{aligned} S_{i,\mu}^{z'} &= S - a_{i,\mu}^{\dagger} a_{i,\mu}, \\ S_{i,\mu}^{+'} &= \sqrt{2S - a_{i,\mu}^{\dagger} a_{i,\mu}} a_{i,\mu} \approx \sqrt{2S} a_{i,\mu}, \\ S_{i,\mu}^{-'} &= a_{i,\mu}^{\dagger} \sqrt{2S - a_{i,\mu}^{\dagger} a_{i,\mu}} \approx \sqrt{2S} a_{i,\mu}^{\dagger}, \end{aligned} \quad (6)$$

i.e. $S^{x'} = \sqrt{S/2}(a + a^{\dagger})$ and $S^{y'} = -i\sqrt{S/2}(a - a^{\dagger})$ at the bilinear level. Lab-frame components recover the Bertaut signs via $S_{i,\mu}^{a,\text{lab}} = \eta_{\mu}^a S_{i,\mu}^{a',\text{loc}}$ for $a = x, y, z$, up to the small canting rotation R_{β_c} that mixes $x \leftrightarrow z$ uniformly across sublattices and only renormalizes coefficients.

B. Bilinear bond Hamiltonian

Consider a generic Fe-Fe bond connecting site $i \in \mu$ and $j \in \nu$ with Heisenberg coupling J_{ij} and DM vector $\mathbf{d}_{ij}^{\text{lab}}$. In local frames

$$\begin{aligned} \mathbf{S}_{i,\mu}^{\text{lab}} \cdot \mathbf{S}_{j,\nu}^{\text{lab}} &= \mathbf{S}_{i,\mu}^{\text{loc}T} (D_{\mu}^T D_{\nu}) \mathbf{S}_{j,\nu}^{\text{loc}} \\ &= \sum_a \eta_{\mu}^a \eta_{\nu}^a S_{i,\mu}^{a'} S_{j,\nu}^{a'}. \end{aligned} \quad (7)$$

Define the bond-resolved relative sign $\sigma_{\mu\nu}^a \equiv \eta_{\mu}^a \eta_{\nu}^a \in \{\pm 1\}$. Then the Heisenberg part of the bond, after substituting (6) and keeping bilinear-in-boson terms (the linear pieces vanish by (4) on the saturated reference), becomes

$$\begin{aligned} H_{ij}^{\text{Heis},(2)} &= J_{ij} [-\sigma_{\mu\nu}^z S(a_i^{\dagger} a_i + a_j^{\dagger} a_j) \\ &\quad + \frac{S}{2} \sigma_{\mu\nu}^+ (a_i^{\dagger} a_j + a_j^{\dagger} a_i) \\ &\quad + \frac{S}{2} \sigma_{\mu\nu}^- (a_i a_j + a_i^{\dagger} a_j^{\dagger})], \end{aligned} \quad (8)$$

where we introduced the symmetric/antisymmetric combinations

$$\sigma_{\mu\nu}^{\pm} \equiv \frac{1}{2} (\sigma_{\mu\nu}^x \pm \sigma_{\mu\nu}^y). \quad (9)$$

Thus $\sigma_{\mu\nu}^z = +1$ (parallel local z , “ferromagnetic bond”) gives a normal hopping with diagonal stiffness, while $\sigma_{\mu\nu}^z = -1$ (antiparallel local z , “antiferromagnetic bond”) gives the standard particle-hole pair $a_i a_j + a_i^{\dagger} a_j^{\dagger}$ structure that produces the Bogoliubov gap. For Pbnm in-plane bonds, $\sigma_{\mu\nu}^z = -1$ for the four nearest neighbors of every Fe.

C. DM bond term and the qAFM gap

The lab-frame DM term $\mathbf{d}_{ij} \cdot \mathbf{S}_{i,\mu} \times \mathbf{S}_{j,\nu}$ becomes, in local frames,

$$\sum_a \tilde{d}_{ij}^a (\mathbf{S}_{i,\mu}^{\text{loc}} \times \mathbf{S}_{j,\nu}^{\text{loc}})^a, \quad \tilde{d}_{ij}^a = \eta_{\mu}^a \eta_{\nu}^a d_{ij}^a \det(R_{\beta_c}), \quad (10)$$

i.e. the cross-product structure is preserved up to a per-component sign $\sigma_{\mu\nu}^a$ (the small canting rotation R_{β_c} also rotates $\tilde{d}^z$ into $\tilde{d}^x$ at $O(\beta_c)$, which is the very rotation that produces F_x from G_z and gives the qFM mode its weak-FM character).

The bilinear-in-boson contribution from (10) reads (using $S^{x'} = \sqrt{S/2}(a + a^{\dagger})$, $S^{y'} = -i\sqrt{S/2}(a - a^{\dagger})$, $S^{z'} =$

$$S - a^\dagger a)$$

$$H_{ij}^{\text{DM},(2)} = -iS \tilde{d}_{ij}^z (a_i^\dagger a_j - a_j^\dagger a_i) + S [\tilde{d}_{ij}^x(\dots) + \tilde{d}_{ij}^y(\dots)], \quad (11)$$

where the $\tilde{d}^{x,y}$ pieces produce the $a^\dagger a^\dagger + aa$ anomalous terms responsible for the DM-induced gap at the qAFM mode [7, 8]. The explicit microscopic sign rule (Fe1 $\rightarrow$ Fe0 in-plane bond carries $(0, +D_1, +D_2)$ while Fe2 $\rightarrow$ Fe3 carries $(0, -D_1, +D_2)$, as in `apply_tmfeo3_fe_sector`) makes the per-bond DM vector *sublattice-pair* dependent, but always reproduces the uniform $+\eta_\mu^x F_x$ canting; in $\mathbf{k}$ -space this translates to a sublattice-off-diagonal hopping matrix whose Bertaut-class (F, G, A, C) decomposition mirrors that of the Heisenberg part.

D. Single-ion anisotropy

The Fe SIA $-K_a S_{i,\mu}^{a,2} = -K_a \eta_\mu^{a,2} S_{i,\mu}^{a',2} = -K_a S_{i,\mu}^{a',2}$ is invariant under η_μ (squares of signs). It bilinearizes to

$$H_i^{\text{SIA},(2)} = -S[(K_x - K_z) + (K_y - K_z)] a_i^\dagger a_i + \frac{S}{2}(K_x - K_y)(a_i^2 + a_i^{\dagger 2}), \quad (12)$$

i.e. an easy-axis (K_z) contribution gives uniform diagonal energy shift, easy-plane biaxiality ($K_x - K_y \neq 0$) gives a local Bogoliubov-pair gap. Using the microscopic diag(K_a, K_b, K_c) implemented in `TmFe03_Fe`, the local-frame substitution is identical because the diagonal anisotropy matrix is sign-invariant.

E. Zeeman

Static lab Zeeman $-\mathbf{h} \cdot \mathbf{S}_{i,\mu}^{\text{lab}}$ becomes $-\sum_a \eta_\mu^a h^a S_{i,\mu}^{a'}$; only the staggered $h^a \eta_\mu^a$ uniform component along local z' contributes at $O(S)$, giving a uniform diagonal shift $h_z^{\text{eff}} = \eta_\mu^z h^z + \beta_c$ -mixed terms.

III. TM SECTOR: GENERALIZED SU(3) FLAVOR-WAVE BOSONS

A. Schwinger three-boson representation

Following Papanicolaou [2] and the multiboson/flavor-wave formulation [3–5] of $\text{SU}(N)$ magnetism, introduce on each Tm site $i \in \mu$ three Schwinger bosons $\{b_{i,\mu}^{(\alpha)}\}_{\alpha=1,2,3}$ associated with the three CEF eigenstates $\{|E_1\rangle, |E_2\rangle, |E_3\rangle\}$, satisfying

$$[b^{(\alpha)}, b^{(\beta)\dagger}] = \delta_{\alpha\beta}, \quad \sum_{\alpha=1}^3 b_{i,\mu}^{(\alpha)\dagger} b_{i,\mu}^{(\alpha)} = M, \quad (13)$$

with $M = 1$ for a single CEF state per Tm site. Any operator $\hat{O}$ acting in the qutrit Hilbert space at site i is

then bilinear in the b 's:

$$\hat{O}_{i,\mu} = \sum_{\alpha\beta=1}^3 \langle E_\alpha | \hat{O} | E_\beta \rangle b_{i,\mu}^{(\alpha)\dagger} b_{i,\mu}^{(\beta)}. \quad (14)$$

In particular the Gell-Mann matrices $\lambda_a = \sum_{\alpha\beta} (\lambda_a)_{\alpha\beta} |E_\alpha\rangle \langle E_\beta|$ become $\lambda_{a,i,\mu} = \sum_{\alpha\beta} (\lambda_a)_{\alpha\beta} b_{i,\mu}^{(\alpha)\dagger} b_{i,\mu}^{(\beta)}$.

B. Condensate and HP-like expansion around $|E_1\rangle$

In the Γ_2 phase the condensed boson is $b^{(1)}$. Eliminate it with the constraint (13):

$$b_{i,\mu}^{(1)} \rightarrow \sqrt{M - b_{i,\mu}^{(2)\dagger} b_{i,\mu}^{(2)} - b_{i,\mu}^{(3)\dagger} b_{i,\mu}^{(3)}} \approx \sqrt{M}, \quad (15)$$

expanded to bilinear order. The two physical flavor bosons of the Tm qutrit are then $b_{i,\mu}^{(2)}$ and $b_{i,\mu}^{(3)}$, i.e. exactly the two CEF excitations $|E_1\rangle \rightarrow |E_2\rangle$ and $|E_1\rangle \rightarrow |E_3\rangle$.

Substituting (15) into (14) gives the linear-in- b representation of the Gell-Mann operators (writing $b^{(\alpha)}$ shorthand and dropping site indices)

$$\lambda_a \simeq M (\lambda_a)_{11} + \sqrt{M} \sum_{\alpha=2,3} [(\lambda_a)_{1\alpha} b^{(\alpha)} + (\lambda_a)_{\alpha 1} b^{(\alpha)\dagger}] + O(b^\dagger b), \quad (16)$$

plus quadratic corrections of the schematic form $\sum_{\alpha\beta \in \{2,3\}} (\lambda_a)_{\alpha\beta} b^{(\alpha)\dagger} b^{(\beta)}$ and the $b^\dagger b$ contraction inside $b^{(1)}$ that renormalizes diagonal pieces.

Concretely, with the basis ordering $(|E_1\rangle, |E_2\rangle, |E_3\rangle)$ and the implementation's indexing $\{\lambda_a\}_{a=1..8}$ in `include/classical_spin/.../gellmann.h`: the linear-in- b pieces are non-zero only for the columns/rows $\alpha = 1$ of each λ_a . The dipole-active triplet $\{\lambda_2, \lambda_5, \lambda_7\}$ and its coherence partners $\{\lambda_1, \lambda_4, \lambda_6\}$ have non-zero $(1, 2), (1, 3), (2, 3)$ matrix elements; the population channels λ_3, λ_8 are diagonal and contribute only at $O(M)$ (mean-field shift) and at $O(b^\dagger b)$ (number renormalizations).

C. CEF Hamiltonian as flavor-boson dispersion

The bare CEF Hamiltonian projected on the three singlets is $H_{\text{CEF}} = \text{diag}(0, e_1, e_2)$ in the eigenbasis (using the $E_1 < E_2 < E_3$ ordering of [12] Sec. III). Via (14) this is exactly

$$H_{i,\mu}^{\text{CEF}} = e_1 b_{i,\mu}^{(2)\dagger} b_{i,\mu}^{(2)} + e_2 b_{i,\mu}^{(3)\dagger} b_{i,\mu}^{(3)}, \quad (17)$$

i.e. the on-site flavor-boson dispersions are the bare CEF gaps. The representation (17) also follows directly from the constant-field encoding $(0, 0, \alpha, 0, 0, 0, \beta)$ used in `apply_tmfeo3_tm_sector`, with $\alpha = e_1$ and $\beta = (2e_2 - e_1)/\sqrt{3}$, by using the matrix elements $(\lambda_3)_{22} = -1, (\lambda_8)_{22} = -1/\sqrt{3}, (\lambda_3)_{33} = 0, (\lambda_8)_{33} = -2/\sqrt{3}$.

D. Tm Zeeman

The Tm Zeeman, $-\sum_a B_a J_a^{\text{phys}}$, where $J_a^{\text{phys}} = \sum_\alpha \mu_{a\alpha} \lambda_{2,5,7}$ on the active triplet (cf. Sec. III B of [12]), becomes after (16)

$$H_{i,\mu}^{Z,\text{Tm},(1)} = -\sqrt{M} \sum_{a;\alpha=2,3} B_a^{\text{loc},\mu} (J_a^{\text{phys}})_{1\alpha} b_{i,\mu}^{(\alpha)} + \text{H.c.},$$

$$B_a^{\text{loc},\mu} = \eta_\mu^a B_a, \quad (18)$$

where the local-frame field $B_a^{\text{loc},\mu} = \eta_\mu^a B_a$ encodes the staggering of the Tm Zeeman field by F_μ . This linear-in- b piece displaces the flavor-boson vacuum: in a true ground state it is re-absorbed by re-defining the condensate $|E_1\rangle \rightarrow |\tilde{E}_1\rangle$ admixing $|E_2\rangle, |E_3\rangle$ at first order in B/e . At the level of the linearized EOM it acts as a coherent driving force, exactly mirroring the THz Zeeman drive of [12] Sec. VI.

E. Tm-Tm bilinear couplings

The diagonal-in- a Tm-Tm bilinears $\mathcal{J}_a \lambda_a^{(i)} \lambda_a^{(j)}$ become, via (16) and using $(\lambda_a)_{11} \neq 0$ only for $a = 3, 8$,

$$H_{ij}^{\text{TmTm},(2)} = M \mathcal{J}_a (\lambda_a)_{11} [(\lambda_a^{(i)})^{(2)} + (\lambda_a^{(j)})^{(2)}]$$

$$+ M \mathcal{J}_a \sum_{\alpha,\beta=2,3} (\lambda_a)_{1\alpha} (\lambda_a)_{\beta 1} b_i^{(\alpha)} b_j^{(\beta)\dagger} + \dots, \quad (19)$$

where $(\lambda_a^{(i)})^{(2)}$ denotes the quadratic-in- b part of $\lambda_a^{(i)}$. The first line renormalizes the on-site flavor dispersion and the second produces inter-site flavor hopping. The sublattice signs for λ_a on the Tm side are read off from F_μ : for the active/coherence indices $\{2, 5, 7, 1, 4, 6\}$ the F_μ block coincides with the SU(2)-induced sign η_μ^a on the (x, y, z) -like components, while the population channels $\{3, 8\}$ are unaffected (F_μ is identity there).

IV. FE-TM COUPLING SECTORS

A. χ exchange (bilinear)

The χ term [12] Sec. V is bilinear and couples the Fe spin to the Tm dipole channels $\{\lambda_2, \lambda_5, \lambda_7\}$:

$$H_\chi = \sum_{(i,j) \in \text{FeTm}} \sum_a \chi_{ij,a} S_{i,\mu}^a S_{j,\mu}^a, \quad (20)$$

with the inversion-related bonds carrying χ_{inv} flipping $\{5, 7\}$ (the A_2^+ partners) and leaving $\{2\}$ (A_1^+) invariant – precisely the rule encoded in `apply_tmfeo3_fe_tm_couplings`.

Linearizing with (6) and (16): the Fe factor gives the linear pieces $S^{x',y'} \sim (a + a^\dagger), (a - a^\dagger)$ at $O(\sqrt{S})$ plus a

classical contribution $\langle S^{z'} \rangle = S$ along local z , while the Tm factor gives linear pieces $\sim (b^{(2)} + b^{(2)\dagger}), (b^{(3)} + b^{(3)\dagger})$ at $O(\sqrt{M})$ plus a vanishing mean-field $\langle \lambda_{2,5,7} \rangle = 0$ on $|E_1\rangle$ for the active triplet. Therefore:

- the mean-field piece $\chi_{ij,a} S \eta_\mu^a \langle \lambda_{j,\nu}^a \rangle = 0$ vanishes ($|E_1\rangle$ is a non-Kramers singlet, no spontaneous dipole), so χ does *not* shift the Tm flavor vacuum at zeroth order; this is the symmetry statement that the Tm “magnetization” is purely Van-Vleck (cf. [12] Sec. V D);
- the cross term $\chi_{ij,a} \langle S_{i,\mu}^a \rangle \lambda_{j,\nu}^a$ acts as a linear-in- b field on the Tm flavor bosons, with strength $\chi_{ij,a} S \eta_\mu^a \delta^{a,z}$ – the Bertaut- F/G pattern selects which Tm sublattice receives which sign;
- the genuine bilinear hybridization $\chi_{ij,a} \sqrt{S} \sqrt{M} (a + a^\dagger)(b^{(\alpha)} + b^{(\alpha)\dagger})$ couples Fe magnons to Tm flavor bosons with the $a \in \{x, y\}$ components of $\mathbf{S}$ pairing with $\lambda_{5,7}$ via $(\lambda_a)_{12,13}$, and similarly for $a = z$.

The inversion-odd bonds (χ_{inv} on partners $\nu = \bar{\mu}$ related by P) carry the opposite sign on $\{5, 7\}$, so when summed over the four bonds connecting a given Tm sublattice they preserve the A_1^+ (λ_2) channel and *cancel* the A_2^+ ($\lambda_{5,7}$) channels at $\mathbf{k} = 0$, leaving a sublattice-staggered hybridization at finite $\mathbf{k}$.

B. W trilinear (Fe-Fe-Tm)

The W term is trilinear:

$$H_W = \sum_{\langle ijk \rangle} \sum_a W_a (S_{i,\mu} \cdot S_{j,\nu}) \lambda_{k,\rho}^a, \quad (21)$$

with even (A_1^+) channels $a \in \{1, 3, 8\}$ surviving inversion and odd (A_2^+) channels $a \in \{4, 6\}$ flipping on the inversion partner. Expand around the classical state by writing $\mathbf{S} = \langle \mathbf{S} \rangle + \delta \mathbf{S}$ and $\lambda^a = \langle \lambda^a \rangle + \delta \lambda^a$:

$$H_W^{(2)} = W_a [\underbrace{\langle \mathbf{S}_i \mathbf{S}_j \rangle \delta \lambda_k^a}_{\text{(i) Tm field renorm.}} + \underbrace{\langle \lambda_k^a \rangle (\delta \mathbf{S}_i \cdot \mathbf{S}_j^{\text{cl}} + \mathbf{S}_i^{\text{cl}} \cdot \delta \mathbf{S}_j)}_{\text{(ii) Fe dispersion renorm.}} + \underbrace{(\delta \mathbf{S}_i \cdot \delta \mathbf{S}_j) \langle \lambda_k^a \rangle}_{\text{(iii) Fe magnon-magnon}} + \underbrace{\langle \mathbf{S}_i \rangle \cdot \delta \mathbf{S}_j \delta \lambda_k^a + (i \leftrightarrow j)}_{\text{(iv) Fe-Tm hybridization}}]. \quad (22)$$

The four pieces have distinct physical content:

- (i) renormalizes the Tm Zeeman / CEF field by the *instantaneous* Fe correlator $\langle \mathbf{S}_i \cdot \mathbf{S}_j \rangle = \sigma_{\mu\nu}^z S^2 + O(1/S)$, giving a sublattice-resolved shift $W_a \sigma_{\mu\nu}^z S^2 \eta_\rho^a$ (only the A_1^+ channels $\{1, 3, 8\}$ contribute on $|E_1\rangle$; the A_2^+ channels $\{4, 6\}$ have $\langle \lambda^a \rangle = 0$);

- (ii) renormalizes the Fe magnon dispersion proportionally to $W_a \langle \lambda_k^a \rangle$, which is non-zero only for the diagonal A_1^+ channels $a \in \{3, 8\}$ on the $|E_1\rangle$ vacuum;
- (iii) is the standard magnon–magnon scattering vertex from a biquadratic term; at the linearized level it contributes $W_a \langle \lambda^a \rangle \delta \mathbf{S} \cdot \delta \mathbf{S}$, i.e. an *additional* Heisenberg bond stiffness, again only for $a \in \{3, 8\}$;
- (iv) is the linear hybridization between an Fe magnon (through $\delta \mathbf{S}_j$) and a Tm flavor boson (through $\delta \lambda_k^a$), weighted by $\langle \mathbf{S}_i \rangle = \eta_\mu^z S \hat{z}'_\mu$ on the partner site. This piece is the genuine W -induced magnon $\leftrightarrow$ flavor-boson mixing; its sign follows the A_1^+/A_2^+ Bertaut pattern of the W channels.

V. FOURIER TRANSFORM ON THE PBNM CELL

A. Definitions

For each sublattice μ introduce

$$a_{\mu, \mathbf{k}} = \frac{1}{\sqrt{N_c}} \sum_{\mathbf{R}} e^{-i\mathbf{k}\mathbf{R}} a_{\mathbf{R}+\mathbf{r}_\mu^{\text{Fe}}},$$

$$b_{\mu, \mathbf{k}}^{(\alpha)} = \frac{1}{\sqrt{N_c}} \sum_{\mathbf{R}} e^{-i\mathbf{k}\mathbf{R}} b_{\mathbf{R}+\mathbf{r}_\mu^{\text{Tm}}}^{(\alpha)}, \quad \alpha = 2, 3, \quad (23)$$

with $\mathbf{R}$ summed over the N_c unit cells and $\mathbf{r}_\mu^{\text{Fe, Tm}}$ the sublattice positions. We *do not* include the sublattice offset in the phase factor: this Fourier convention puts all sublattice phase structure in the dispersion matrix and keeps $H(\mathbf{k})$ periodic over the Pbnm Brillouin zone (sublattice-orbital gauge).

B. Bond geometric factors

For each Fe–Fe orbit $\{(\mu_0, \nu_0) : \mathbf{d}_0\}$ let $\mathbf{d}_p^{(\mu\nu)}$ ($p = 1 \dots n_{\mu\nu}$) be all bonds in that orbit connecting Fe sublattices μ, ν . The geometric structure factor is

$$\gamma_{\mu\nu}^{\text{FeFe}}(\mathbf{k}) = \sum_p e^{i\mathbf{k}\mathbf{d}_p^{(\mu\nu)}}, \quad \gamma_{\nu\mu}^{\text{FeFe}}(\mathbf{k}) = \overline{\gamma_{\mu\nu}^{\text{FeFe}}(\mathbf{k})}. \quad (24)$$

Analogously for Fe–Tm ($\gamma_{\mu\nu}^{\text{FeTm}}$) and Tm–Tm ($\gamma_{\mu\nu}^{\text{TmTm}}$) orbits. Note that in Pbnm the 16 Fe–Tm bonds split into 4 orbits of 4 bonds each, and the four bonds within an orbit are related by Pbnm symmetry operations – this is exactly the bond list encoded in `apply_tmfeo3_fe_tm_couplings`.

C. Bosonic Nambu vector

Define the 24-component Nambu vector

$$\Psi_{\mathbf{k}} = \left(\underbrace{a_{0, \mathbf{k}}, \dots, a_{3, \mathbf{k}}}_{\text{Fe magnons}}, \underbrace{a_{0, -\mathbf{k}}^\dagger, \dots, a_{3, -\mathbf{k}}^\dagger}_{\text{Fe Nambu}}, \mathbf{b}_{\mathbf{k}}^{(2)}; \mathbf{b}_{-\mathbf{k}}^{(2)\dagger}; \mathbf{b}_{\mathbf{k}}^{(3)}; \mathbf{b}_{-\mathbf{k}}^{(3)\dagger} \right)^T, \quad (25)$$

where each Tm block $\mathbf{b}_{\mathbf{k}}^{(\alpha)} = (b_{0, \mathbf{k}}^{(\alpha)}, \dots, b_{3, \mathbf{k}}^{(\alpha)})$ is a 4-vector over the four Tm sublattices. The bosonic commutator metric is the Bogoliubov τ_3

$$\tau_3 = \text{diag}(+I_4, -I_4, +I_4, -I_4, +I_4, -I_4). \quad (26)$$

VI. LINEARIZED HAMILTONIAN IN $\mathbf{k}$ -SPACE

A. Block structure

Collecting Sections II–IV the quadratic Hamiltonian is

$$H^{(2)} = \frac{1}{2} \sum_{\mathbf{k}} \Psi_{\mathbf{k}}^\dagger H(\mathbf{k}) \Psi_{\mathbf{k}} + \text{const.}, \quad (27)$$

with the block matrix

$$H(\mathbf{k}) = \begin{pmatrix} H_{\text{FeFe}}(\mathbf{k}) & H_{\text{FeTm}}^{(2)}(\mathbf{k}) & H_{\text{FeTm}}^{(3)}(\mathbf{k}) \\ H_{\text{FeTm}}^{(2)\dagger}(\mathbf{k}) & H_{\text{TmTm}}^{(2)}(\mathbf{k}) & V^{(23)}(\mathbf{k}) \\ H_{\text{FeTm}}^{(3)\dagger}(\mathbf{k}) & V^{(23)\dagger}(\mathbf{k}) & H_{\text{TmTm}}^{(3)}(\mathbf{k}) \end{pmatrix}. \quad (28)$$

Each top-level block is itself 2×2 in Nambu (normal / anomalous) and 4×4 in sublattice. Explicit entries follow.

B. Fe magnon block (8×8)

In the Nambu basis $(a_\mu, a_\mu^\dagger)$ the Fe block reads

$$H_{\text{FeFe}}(\mathbf{k}) = \begin{pmatrix} A_{\text{Fe}}(\mathbf{k}) & B_{\text{Fe}}(\mathbf{k}) \\ B_{\text{Fe}}^\dagger(\mathbf{k}) & A_{\text{Fe}}^T(-\mathbf{k}) \end{pmatrix}, \quad (29)$$

with

$$[A_{\text{Fe}}(\mathbf{k})]_{\mu\nu} = \delta_{\mu\nu} \Omega_\mu^{\text{Fe}} + \frac{S}{2} \sigma_{\mu\nu}^+ J_{\mu\nu} \gamma_{\mu\nu}^{\text{FeFe}}(\mathbf{k}) - iS \sigma_{\mu\nu}^z \tilde{d}_{\mu\nu}^z \gamma_{\mu\nu}^{\text{FeFe}}(\mathbf{k}), \quad (30)$$

$$[B_{\text{Fe}}(\mathbf{k})]_{\mu\nu} = \frac{S}{2} \sigma_{\mu\nu}^- J_{\mu\nu} \gamma_{\mu\nu}^{\text{FeFe}}(\mathbf{k}) + \frac{S}{2} \tilde{d}_{\mu\nu}^\perp \gamma_{\mu\nu}^{\text{FeFe}}(\mathbf{k}) + \delta_{\mu\nu} \frac{S}{2} (K_{x'} - K_{y'}), \quad (31)$$

where Ω_μ^{Fe} collects the on-site terms (sum of bond $-\sigma_{\mu\nu}^z JS$ stiffnesses, SIA diagonal $-S(K_{x'} + K_{y'} - 2K_{z'})$, Zeeman $\eta_\mu^z h^z$, and the W -induced renormalization of (ii) and (iii) in (22)), and $\tilde{d}^\perp$ contains the in-local-plane DM components responsible for the qAFM gap. With the microscopic values of `apply_tmfeo3_fe_sector` the in-plane DM matrix elements that mix Fe1 $\leftrightarrow$ Fe0 and Fe2 $\leftrightarrow$ Fe3 carry the explicit sign rule $(0, +D_1, +D_2)$ vs. $(0, -D_1, +D_2)$ encoded in $\tilde{d}_{\mu\nu}$.

C. Tm flavor-boson blocks (8×8 each)

For flavor $\alpha \in \{2, 3\}$,

$$H_{\text{TmTm}}^{(\alpha)}(\mathbf{k}) = \begin{pmatrix} A_{\text{Tm}}^{(\alpha)}(\mathbf{k}) & B_{\text{Tm}}^{(\alpha)}(\mathbf{k}) \\ B_{\text{Tm}}^{(\alpha)\dagger}(\mathbf{k}) & A_{\text{Tm}}^{(\alpha)T}(-\mathbf{k}) \end{pmatrix}, \quad (32)$$

with

$$[A_{\text{Tm}}^{(\alpha)}(\mathbf{k})]_{\mu\nu} = \delta_{\mu\nu} (e_{\alpha-1} + \Delta\Omega_{\mu}^{(\alpha)}) + \sum_a \mathcal{J}_a \kappa_{a;\mu\nu}^{(\alpha\alpha)} \gamma_{\mu\nu}^{\text{TmTm}}(\mathbf{k}), \quad (33)$$

$$[B_{\text{Tm}}^{(\alpha)}(\mathbf{k})]_{\mu\nu} = \sum_a \mathcal{J}_a \tilde{\kappa}_{a;\mu\nu}^{(\alpha\alpha)} \gamma_{\mu\nu}^{\text{TmTm}}(\mathbf{k}), \quad (34)$$

where $e_1 = \Delta_{12}, e_2 = \Delta_{13}$ are the bare CEF gaps, $\Delta\Omega_{\mu}^{(\alpha)}$ is the renormalization from W piece (i) of (22) and from the Tm–Tm self-correlator $M\mathcal{J}_a(\lambda_a)_{11}$, and the bond coefficients

$$\kappa_{a;\mu\nu}^{(\alpha\beta)} = (F_{\mu}\lambda_a F_{\mu}^{-1})_{1\alpha} \cdot (F_{\nu}\lambda_a F_{\nu}^{-1})_{\beta 1}, \quad (35)$$

$\tilde{\kappa}_{a;\mu\nu}^{(\alpha\beta)} = (F_{\mu}\lambda_a F_{\mu}^{-1})_{1\alpha} \cdot (F_{\nu}\lambda_a F_{\nu}^{-1})_{1\beta}$ (anomalous) encode the Bertaut sublattice signs of the Tm side. The flavor-off-diagonal hybridization $V^{(23)}(\mathbf{k})$ has the same form with $\beta=3, \alpha=2$.

D. Fe–Tm hybridization (8×8)

The χ piece, plus the W -piece (iv), yield

$$H_{\text{FeTm}}^{(\alpha)}(\mathbf{k}) = \begin{pmatrix} U^{(\alpha)}(\mathbf{k}) & V^{(\alpha)}(\mathbf{k}) \\ V^{(\alpha)*}(-\mathbf{k}) & U^{(\alpha)*}(-\mathbf{k}) \end{pmatrix}, \quad (36)$$

with

$$[U^{(\alpha)}(\mathbf{k})]_{\mu\nu} = \sqrt{SM} \sum_a \left[\chi_a^{(\mu\nu)} + W_a^{(\mu\nu)} S \right] \eta_{\mu}^a \Lambda_{a;\nu}^{(\alpha)} \gamma_{\mu\nu}^{\text{FeTm}}(\mathbf{k}),$$

$$\Lambda_{a;\nu}^{(\alpha)} \equiv (F_{\nu}\lambda_a F_{\nu}^{-1})_{1\alpha}, \quad (37)$$

$$[V^{(\alpha)}(\mathbf{k})]_{\mu\nu} = \sqrt{SM} \sum_a \left[\chi_a^{(\mu\nu)} + W_a^{(\mu\nu)} S \right] \eta_{\mu}^a \bar{\Lambda}_{a;\nu}^{(\alpha)} \gamma_{\mu\nu}^{\text{FeTm}}(\mathbf{k}),$$

$$\bar{\Lambda}_{a;\nu}^{(\alpha)} \equiv (F_{\nu}\lambda_a F_{\nu}^{-1})_{\alpha 1}, \quad (38)$$

where $\chi_a^{(\mu\nu)}$ and $W_a^{(\mu\nu)}$ are the bond-resolved coupling constants (carrying the A_1^+/A_2^+ sign-flip rule for inversion partners), and the η_{μ}^a factor is the Fe local-frame sign on component a . The four $\mathbf{k}$ -dependent structure factors $\gamma_{\mu\nu}^{\text{FeTm}}(\mathbf{k})$ encode the four bonds within each orbit.

VII. LINEARIZED EQUATIONS OF MOTION

A. Paraunitary EOM

The Heisenberg EOM $i\partial_t \mathcal{O} = [\mathcal{O}, H^{(2)}]$, applied to each component of $\Psi_{\mathbf{k}}$ in (25) and using the bosonic com-

mutators encoded by τ_3 in (26), gives the linear EOM

$$i\partial_t \Psi_{\mathbf{k}}(t) = \tau_3 H(\mathbf{k}) \Psi_{\mathbf{k}}(t). \quad (39)$$

The normal modes $\omega_n(\mathbf{k})$ are the positive eigenvalues of $\tau_3 H(\mathbf{k})$ obtained by paraunitary (Colpa) diagonalization [6]: find $T(\mathbf{k})$ such that $T^{\dagger} H T = \text{diag}(\omega_n, \omega_n)$ and $T^{\dagger} \tau_3 T = \tau_3$. The resulting eigenvalues split into eight magnon bands (4 sublattices $\times$ 2 Nambu, but the bosonic symmetry $H^*(-\mathbf{k}) = H(\mathbf{k})$ identifies the upper-half with the lower-half) and eight flavor-boson bands (two flavors $\alpha = 2, 3$ over four sublattices), with the four hybridization terms $H_{\text{FeTm}}^{(\alpha)}, V^{(23)}$ mixing all of them. At $\mathbf{k} = 0$ the four-fold sublattice degeneracy of the Heisenberg part lifts into the Bertaut $\{F, G, A, C\}$ irreps – recovering the qFM and qAFM modes of Sec. VIII of [12] as the two Goldstone-like combinations of the four Fe sublattices.

B. Symmetry constraints

Hermiticity and bosonic statistics impose

$$H(\mathbf{k}) = H(\mathbf{k})^{\dagger}, \quad (40)$$

$$H(-\mathbf{k}) = \Sigma_1 H^*(\mathbf{k}) \Sigma_1, \quad \Sigma_1 = \sigma_1 \otimes I_{12}, \quad (41)$$

where σ_1 swaps the Nambu doubles. The Bertaut symmetry manifests as block-diagonalization at the high-symmetry momenta $\mathbf{k} = \Gamma$ and $\mathbf{k} = (0, 0, \pi)$ where the four-sublattice quadrant is the irreducible representation matrix of the Klein four group of (2).

C. Recovering the $\mathbf{k}=0$ joint EOM

At $\mathbf{k}=0$, projecting (39) onto the Bertaut F_x and G_z modes (i.e. the qFM and qAFM linear combinations) and on the two condensed Tm flavor modes reproduces the joint 4+2 component linear EOM of [12] Sec. XI exactly. Concretely, the qFM and qAFM eigen-frequencies are

$$\omega_{\text{qFM}}^2 = 2S K_z (2J_{\text{tot}} + K_x) + (g\mu_B H)^2, \quad \omega_{\text{qAFM}}^2 = 2S (K_x - K_z) (2J_{\text{tot}}) + (D_{\text{DM}})^2 + (\text{Tm renorm.}), \quad (42)$$

in agreement with the Herrmann formulas [7, 8], and the Tm flavor frequencies are $\omega_{\text{Tm}}^{(\alpha)}(\mathbf{k}=0) = e_{\alpha-1} + \Delta\Omega^{(\alpha)}$, i.e. the bare CEF gaps with W -induced shifts – exactly the form used in Sec. VII of [12] for linearization around $|E_1\rangle$.

VIII. BERTAUT DIAGONALIZATION: EXPLICIT QFM, QAFM, AND TM FLAVOR OPERATORS

The four-sublattice HP and flavor bosons of Sec. II–III are not yet normal modes: the Γ -point quadratic kernel

(28) is block-diagonalized by the Klein-four unitary that implements the Bertaut decomposition $\{F, G, C, A\}$. We give the change of basis explicitly, then write the two-component Bogoliubov pairs that define the qFM and qAFM magnon creation operators.

A. Klein-four unitary on the four sublattices

The Pbnm sign matrix (2) has rows $(\eta_\mu^x, \eta_\mu^y, \eta_\mu^z) = (+++, +--, -+-, ---)$. The orthogonal 4×4 matrix that diagonalizes its action on the four sublattices is

$$U_B = \frac{1}{2} \begin{pmatrix} +1 & +1 & +1 & +1 \\ +1 & -1 & +1 & -1 \\ +1 & +1 & -1 & -1 \\ +1 & -1 & -1 & +1 \end{pmatrix}, \quad U_B^T U_B = I_4, \quad (43)$$

with the four rows labelled by the Bertaut modes $X \in \{F, G, C, A\}$ in the order corresponding to the staggering patterns $(++++)$ (F, uniform), $(+--+)$ (G, y -component Neel), $(+-+)$ (C, x -component), and $(+--+)$ (A, z -component). Equivalently, denoting the sublattice indices by $\mu \in \{0, 1, 2, 3\}$,

$$[U_B]_{X\mu} = \frac{1}{2} \eta_\mu^X, \quad \eta_\mu^F = +1, \quad \eta_\mu^G = \eta_\mu^y, \quad \eta_\mu^C = \eta_\mu^x, \quad \eta_\mu^A = \eta_\mu^z. \quad (44)$$

Crucially, the four columns of η (2) reproduce the G, C, A rows of U_B – i.e. the Cartesian components x, y, z of any vector quantity *exactly* pick out one Bertaut mode each. This is what produces the celebrated orthoferrite irreps $\Gamma_1 = A_x G_y C_z$, $\Gamma_2 = F_x C_y G_z$, $\Gamma_4 = G_x A_y F_z$, and is the reason qFM and qAFM modes are combinations of *at most two* Bertaut components.

B. Fe Bertaut magnon operators

Define the Bertaut-mode bosons by the same unitary acting on the four sublattice HP operators (we work at $\mathbf{k}=0$; the $\mathbf{k}$ -dependent generalization carries the unitary into a 4×4 rotation within each $\mathbf{k}$ -slab):

$$a_X(\mathbf{k}) = \frac{1}{2} \sum_\mu \eta_\mu^X a_{\mu, \mathbf{k}}, \quad X \in \{F, G, C, A\}, \quad (45)$$

with $[a_X, a_{X'}^\dagger] = \delta_{XX'}$. Their inverse is $a_{\mu, \mathbf{k}} = \frac{1}{2} \sum_X \eta_\mu^X a_X(\mathbf{k})$.

In terms of these, the Fe local-frame spin operators acquire a *single* Bertaut label per Cartesian component:

$$\begin{aligned} \frac{1}{\sqrt{2SN_c}} \sum_\mu S_{\mu, \mathbf{k}}^{x', \text{loc}} &\longrightarrow a_F^\dagger(\mathbf{k}) + a_F(\mathbf{k}) \quad (\text{F-mode quadrature}), \\ \frac{1}{\sqrt{2SN_c}} \sum_\mu S_{\mu, \mathbf{k}}^{y', \text{loc}} &\longrightarrow -i[a_F^\dagger(\mathbf{k}) - a_F(\mathbf{k})], \\ \frac{1}{\sqrt{2SN_c}} \sum_\mu \eta_\mu^y S_{\mu, \mathbf{k}}^{x', \text{loc}} &\longrightarrow a_G^\dagger(\mathbf{k}) + a_G(\mathbf{k}), \text{ etc.} \end{aligned} \quad (46)$$

Combined with the Bertaut tables of [12] Sec. IV C, the *lab-frame* Bertaut order parameters are $F^x = \eta_\mu^x \hat{\eta}_\mu^F S_\mu^{x', \text{loc}}$ etc., so in the Γ_2 phase the qFM mode (small ferromagnetic moment along $\hat{a}$) is built from a_F , and the qAFM mode (Neel order along $\hat{c}$) is built from a_A .

C. Reduced Bertaut kernel at $\mathbf{k}=0$

Substituting (45) into the Fe block (29) and using $\sum_\mu \eta_\mu^X \eta_\mu^{X'} = 4\delta_{XX'}$ together with $\sum_{\mu\nu} \eta_\mu^X \eta_\nu^{X'} J_{\mu\nu} \sigma_{\mu\nu}^{+/-/z} = 4\mathcal{J}_{XX'}^{+/-/z}$ (a 4×4 Bertaut-class matrix), the Fe 8×8 kernel at $\mathbf{k} = 0$ splits into four *decoupled* 2×2 Bogoliubov blocks, one per Bertaut class:

$$H_{\mathbf{k}=0}^{\text{Fe}} = \sum_{X=F,G,C,A} \frac{1}{2} (a_X^\dagger, a_X) \begin{pmatrix} \mathcal{A}_X & \mathcal{B}_X \\ \mathcal{B}_X^* & \mathcal{A}_X \end{pmatrix} \begin{pmatrix} a_X \\ a_X^\dagger \end{pmatrix}, \quad (47)$$

with diagonal stiffness

$$\mathcal{A}_X = S[-zJ_{\text{tot}}\sigma_{XX}^z + 2K_{\text{eff}}^X + g\mu_B H \eta_z^X], \quad (48)$$

$$\mathcal{B}_X = S(K_{x'} - K_{y'})\delta_F^X + S\sigma_{XX}^- J_{\text{tot}} + iS\tilde{d}_{XX}^\perp, \quad (49)$$

where $z=6$ is the Fe coordination, K_{eff}^X groups the SIA contributions diagonal in X , and the $\sigma_{XX}^z, \sigma_{XX}^\pm$ factors are read off from $\sigma_{\mu\nu}^a$ via $\sigma_{XX}^a = \frac{1}{4} \sum_{\mu\nu} \eta_\mu^X \eta_\nu^X \sigma_{\mu\nu}^a$.

D. qFM and qAFM Bogoliubov rotation

Each 2×2 block (47) is diagonalized by the standard $\text{SU}(1,1)$ Bogoliubov rotation,

$$a_X = u_X \alpha_X - v_X^* \alpha_X^\dagger, \quad u_X^2 - |v_X|^2 = 1, \quad (50)$$

with $u_X = \cosh \theta_X$, $v_X = e^{i\phi_X} \sinh \theta_X$ and

$$\tanh 2\theta_X = \frac{|\mathcal{B}_X|}{\mathcal{A}_X}, \quad \omega_X = \sqrt{\mathcal{A}_X^2 - |\mathcal{B}_X|^2}. \quad (51)$$

We identify the canonical magnon operators of orthoferrite physics:

$$\boxed{\alpha_{\text{qFM}} \equiv \alpha_F, \quad \alpha_{\text{qAFM}} \equiv \alpha_A,} \quad (52)$$

with frequencies

$$\begin{aligned} \omega_{\text{qFM}} &= \sqrt{\mathcal{A}_F^2 - |\mathcal{B}_F|^2}, \\ \omega_{\text{qAFM}} &= \sqrt{\mathcal{A}_A^2 - |\mathcal{B}_A|^2}, \end{aligned} \quad (53)$$

which reproduce (42) after substituting the Herrmann coefficients. The G and C Bertaut modes are gapped silent modes (they would become qFM/qAFM in the Γ_1, Γ_4 phases).

E. Tm flavor Bertaut bosons

Apply the same Klein-four unitary to each Tm flavor $\alpha \in \{2, 3\}$,

$$b_X^{(\alpha)}(\mathbf{k}) = \frac{1}{2} \sum_{\mu} \eta_{\mu}^X b_{\mu, \mathbf{k}}^{(\alpha)}, \quad X \in \{F, G, C, A\}. \quad (54)$$

Because the bare CEF Hamiltonian (17) is on-site ($\delta_{\mu\nu}$), the Tm 8×8 block per flavor reduces at $\mathbf{k} = 0$ to four identical copies of a 1×1 "block" (a 2×2 Nambu pair with $\mathcal{B} = 0$):

$$\omega_X^{(\alpha)} = e_{\alpha-1} + \Delta\Omega_X^{(\alpha)}, \quad X \in \{F, G, C, A\}, \quad (55)$$

with $\Delta\Omega_X^{(\alpha)}$ the W - and $\mathcal{J}$ -induced Bertaut splitting from (22)(i) and (19). The splitting follows the standard pattern: A_1^+ Tm–Tm bilinears ($a \in \{1, 3, 8\}$) preserve all four Bertaut classes equally; A_2^+ channels ($a \in \{4, 6\}$) shift A, C relative to F, G .

The four *physical* Tm CEF magnons (which we will see couple to qFM and qAFM) are

$$\beta_X^{(\alpha)} \equiv b_X^{(\alpha)}, \quad \alpha \in \{2, 3\}, \quad X \in \{F, G, C, A\}. \quad (56)$$

Note there is *no* Bogoliubov rotation needed for the Tm flavor bosons in the absence of Tm–Tm anomalous terms (which would require the bilinear $b^{(\alpha)}b^{(\alpha)}$ to be generated, and the bare on-site H_{CEF} has no such term).

F. Selection rules: which Bertaut classes mix?

The Fe–Tm hybridization (36) couples Fe Bertaut mode X to Tm flavor Bertaut mode X' *only* when the bond-resolved coupling $\chi_a^{(\mu\nu)} \eta_{\mu}^a$ has non-vanishing projection on $\eta_{\mu}^X \eta_{\nu}^{X'}$. Using the χ_{inv} sign-flip rule for $\{\lambda_5, \lambda_7\}$ (A_2^+) of `apply_tmfeo3_fe_tm_couplings` the surviving channels at $\mathbf{k} = 0$ are

Fe Bertaut X	Tm flavor Bertaut X'
F (qFM)	F, A
A (qAFM)	F, A
G	G, C
C	G, C

(57)

i.e. qFM and qAFM *both* couple to the same Tm flavor Bertaut subspace $\{F, A\}$, producing a 4×4 hybridization at $\mathbf{k} = 0$ in each flavor channel.

IX. TORQUE EQUATIONS OF MOTION

We now write the EOM of every canonical coordinate, separating the linear (first-order in operators) piece from the genuinely second-order (cubic vertex) magnon-magnon and magnon-flavor pieces. We work with the

Bertaut bosons of (45) and (54) and adopt the canonical quadratures

$$\begin{aligned} \mathcal{Q}_X &= \frac{1}{\sqrt{2}}(a_X^\dagger + a_X), & \mathcal{P}_X &= \frac{i}{\sqrt{2}}(a_X^\dagger - a_X), \\ \mathcal{X}_X^{(\alpha)} &= \frac{1}{\sqrt{2}}(b_X^{(\alpha)\dagger} + b_X^{(\alpha)}), & \mathcal{Y}_X^{(\alpha)} &= \frac{i}{\sqrt{2}}(b_X^{(\alpha)\dagger} - b_X^{(\alpha)}), \end{aligned} \quad (58)$$

with $[\mathcal{Q}_X, \mathcal{P}_{X'}] = i\delta_{XX'}$, etc. We set $\hbar = 1$.

A. First-order (linear) torques: Fe qFM and qAFM

For each Bertaut Fe block (47) the Heisenberg EOM $\dot{\mathcal{O}} = -i[\mathcal{O}, H^{(2)}]$ gives

$$\dot{\mathcal{Q}}_X = (\mathcal{A}_X - \text{Re } \mathcal{B}_X) \mathcal{P}_X - \text{Im } \mathcal{B}_X \mathcal{Q}_X, \quad (59)$$

$$\dot{\mathcal{P}}_X = -(\mathcal{A}_X + \text{Re } \mathcal{B}_X) \mathcal{Q}_X - \text{Im } \mathcal{B}_X \mathcal{P}_X, \quad (60)$$

giving the harmonic-oscillator form $\ddot{\mathcal{Q}}_X + \omega_X^2 \mathcal{Q}_X = 0$ with $\omega_X^2 = \mathcal{A}_X^2 - |\mathcal{B}_X|^2$ for $X \in \{F, A\}$, i.e. the qFM and qAFM modes (53). Equivalent ladder form:

$$\dot{a}_X = -i\mathcal{A}_X a_X - i\mathcal{B}_X a_X^\dagger, \quad \dot{a}_X^\dagger = +i\mathcal{A}_X a_X^\dagger + i\mathcal{B}_X^* a_X. \quad (61)$$

This is the linear torque equation – the bosonic analog of the Landau–Lifshitz torque $\dot{\mathbf{S}} = \mathbf{S} \times \mathbf{h}^{\text{eff}}$ on the Γ_2 ground state, evaluated by linearizing in $\delta\mathbf{S}$ and projecting on the qFM and qAFM eigen-axes.

B. First-order torques: Tm flavor bosons

For the Tm flavor Bertaut bosons (54) with on-site Hamiltonian (55),

$$\dot{\mathcal{X}}_X^{(\alpha)} = \omega_X^{(\alpha)} \mathcal{Y}_X^{(\alpha)}, \quad \dot{\mathcal{Y}}_X^{(\alpha)} = -\omega_X^{(\alpha)} \mathcal{X}_X^{(\alpha)} + f_X^{(\alpha)}(t), \quad (62)$$

where the inhomogeneous driving term $f_X^{(\alpha)}(t)$ collects (a) the THz Zeeman drive (Sec. III Eq. (18)), projected onto Bertaut class X via $\sum_{\mu} \eta_{\mu}^X B_a \eta_{\mu}^a$, and (b) any classical-field source from the Fe vacuum that survives at zeroth order.

C. First-order Fe–Tm hybridization terms

The Fe–Tm hybridization (36), after Bertaut decomposition, contributes a *linear* coupling between $\mathcal{Q}_X$ (or $\mathcal{P}_X$) and $\mathcal{X}_{X'}^{(\alpha)}$ subject to the selection rule (57). Adding

it to (59)–(62):

$$\dot{a}_X = -i\mathcal{A}_X a_X - i\mathcal{B}_X a_X^\dagger - i \sum_{\alpha=2,3} \sum_{X'} \left[g_{XX'}^{(\alpha)} b_{X'}^{(\alpha)} + \tilde{g}_{XX'}^{(\alpha)} b_{X'}^{(\alpha)\dagger} \right], \quad (63)$$

$$\dot{b}_{X'}^{(\alpha)} = -i\omega_{X'}^{(\alpha)} b_{X'}^{(\alpha)} - i \sum_X \left[g_{XX'}^{(\alpha)*} a_X + \tilde{g}_{XX'}^{(\alpha)} a_X^\dagger \right], \quad (64)$$

with vertex coefficients

$$g_{XX'}^{(\alpha)} = \frac{1}{4} \sum_{\mu\nu} \eta_\mu^X \eta_\nu^{X'} [U^{(\alpha)}(\mathbf{k}=0)]_{\mu\nu}, \quad \tilde{g}_{XX'}^{(\alpha)} = \frac{1}{4} \sum_{\mu\nu} \eta_\mu^X \eta_\nu^{X'} [V^{(\alpha)}(\mathbf{k}=0)]_{\mu\nu} \quad (65)$$

read off from (37)–(38). These are the linear torques produced by the static χ and W Fe–Tm bonds.

The complete first-order joint EOM in the canonical-quadrature form is then

$$\dot{\Phi}_{\mathbf{k}} = J \nabla_{\Phi} H^{(2)}, \quad \Phi = (\mathcal{Q}_X, \mathcal{P}_X, \mathcal{X}_X^{(\alpha)}, \mathcal{Y}_X^{(\alpha)}), \quad (66)$$

with the symplectic form $J = \bigoplus (i\sigma_y)$ acting on each $(\mathcal{Q}, \mathcal{P})$ and $(\mathcal{X}, \mathcal{Y})$ pair, and $H^{(2)}$ is the quadratic form (27).

D. Second-order (cubic-vertex) torques

The HP expansion (6) is exact at all orders; truncating it to $S^{x',y'} \propto \sqrt{2S}a$ produces only the bilinear $H^{(2)}$. The next-order pieces come from (i) the $\sqrt{1-a^\dagger a}/(2S)$ correction to $S^\pm$ and (ii) the constraint expansion of the Tm condensate (15), both expanded to one extra order in $1/\sqrt{S}$ and $1/\sqrt{M}$. The leading cubic interactions are:

a. Fe magnon-magnon (3-leg vertex). From the DM bond (10) expanded with $S^{z'} \rightarrow S - a^\dagger a$ and $S^\pm \rightarrow \sqrt{2S}a^{(\dagger)}(1 - a^\dagger a/4S)$:

$$H_{\text{FeFe}}^{(3)} = \sqrt{\frac{S}{2}} \tilde{d}_{ij}^z (a_i^\dagger a_j^\dagger a_j + a_i^\dagger a_i a_j + \text{H.c.}) + \dots, \quad (67)$$

which after Bertaut decomposition becomes

$$H_{\text{FeFe}}^{(3)} = \sum_{X_1 X_2 X_3} V_{X_1 X_2 X_3}^{(3)\text{Fe}} a_{X_1}^\dagger a_{X_2} a_{X_3} + \text{H.c.}, \quad (68)$$

with the four-sublattice sum $\sum_{X_1 X_2 X_3} V_{X_1 X_2 X_3}^{(3)\text{Fe}} \propto \frac{1}{8} \sum_{\mu\nu} \eta_\mu^{X_1} \eta_\nu^{X_2} \eta_\nu^{X_3} \tilde{d}_{\mu\nu}^z / \sqrt{2S}$. The selection rule $\sum_\mu \eta_\mu^{X_1} \eta_\mu^{X_2} \eta_\mu^{X_3} \eta_\mu^{X_4} = 4\delta_{X_1 \oplus X_2, X_3 \oplus X_4}$ (Klein-four addition) ensures that only Bertaut-class-conserving vertices ($X_1 \oplus X_2 \oplus X_3 = F$) survive.

b. Tm flavor-flavor (3-leg vertex). From (16) expanded one more order with $b^{(1)} \rightarrow \sqrt{M} - \frac{1}{2\sqrt{M}}(b^{(2)\dagger} b^{(2)} + b^{(3)\dagger} b^{(3)})$:

$$H_{\text{Tm}}^{(3)} = \frac{1}{\sqrt{M}} \sum_{a, \mathcal{J}_a, \alpha\beta\gamma \in \{2,3\}} T_{a;\alpha\beta\gamma}^{(3)\text{Tm}} b^{(\alpha)\dagger} b^{(\beta)\dagger} b^{(\gamma)} + \text{H.c.}, \quad (69)$$

with coefficients fixed by the matrix elements $(\lambda_a)_{1\beta}(\lambda_a)_{\alpha 1}$ – this is the standard multiboson 3-magnon vertex of [3, 4].

c. Fe–Tm cubic vertex ($1/\sqrt{S}$ from χ , $1/S$ from W). The χ exchange yields a 3-leg Fe-magnon + Tm-flavor + Fe-magnon vertex

$$H_\chi^{(3)} = \frac{1}{\sqrt{2S}} \chi_a^{(\mu\nu)} \eta_\mu^a a_{i,\mu}^\dagger a_{i,\mu} [(\lambda_a)_{12} b_{j,\nu}^{(2)} + (\lambda_a)_{13} b_{j,\nu}^{(3)} + \text{H.c.}] \quad (70)$$

(from the $S^{z'} \rightarrow S - a^\dagger a$ correction acting on the condensate piece of the Fe spin). The W trilinear similarly generates a $\sqrt{S}$ -suppressed Fe-Fe-magnon + Tm-flavor coupling and a $1/\sqrt{M}$ correction that couples Fe magnons to two Tm flavor bosons. All four vertex types have explicit Bertaut form after applying (45) and (54).

E. Compact second-order torque equations

Including the cubic vertices, the EOM in the operator (ladder) basis reads

$$\begin{aligned} \dot{a}_X &= -i[\mathcal{A}_X a_X + \mathcal{B}_X a_X^\dagger] \\ &\quad - i \sum_{\alpha, X'} [g_{XX'}^{(\alpha)} b_{X'}^{(\alpha)} + \tilde{g}_{XX'}^{(\alpha)} b_{X'}^{(\alpha)\dagger}] \\ &\quad - i \sum_{X_1 X_2} [V_{XX_1 X_2}^{(3)\text{Fe}} + V_{X_1 X X_2}^{(3)\text{Fe}*}] a_{X_1}^\dagger a_{X_2} \\ &\quad - i \sum_{\alpha\beta; X_1 X_2} W_{XX_1 X_2; \alpha\beta}^{(3)\text{FeTm}} a_{X_1} b_{X_2}^{(\beta)\dagger} b_{X_2}^{(\alpha)} + \dots, \end{aligned} \quad (71)$$

$$\begin{aligned} \dot{b}_X^{(\alpha)} &= -i\omega_X^{(\alpha)} b_X^{(\alpha)} - i \sum_{X'} [g_{X'X}^{(\alpha)*} a_{X'} + \tilde{g}_{X'X}^{(\alpha)} a_{X'}^\dagger] \\ &\quad - i \sum_{\beta\gamma; Y_1 Y_2} T_{\alpha\beta\gamma; X Y_1 Y_2}^{(3)\text{Tm}} b_{Y_1}^{(\beta)\dagger} b_{Y_2}^{(\gamma)} \\ &\quad - i \sum_{X_1 X_2} U_{XX_1 X_2; \alpha}^{(3)\text{FeTm}} a_{X_1}^\dagger a_{X_2} + \dots, \end{aligned} \quad (72)$$

with the same Bertaut Klein-four selection rule applying to every vertex.

Casting back to canonical quadratures (58), the second-order torque on, e.g., the qFM coordinate is

$$\ddot{\mathcal{Q}}_{\text{qFM}} + \omega_{\text{qFM}}^2 \mathcal{Q}_{\text{qFM}} = \mathcal{F}_{\text{qFM}}^{\text{hyb}}(t) + \mathcal{F}_{\text{qFM}}^{\text{m-m}}[\mathcal{Q}, \mathcal{X}] + \mathcal{F}_{\text{qFM}}^{\text{drv}}(t), \quad (73)$$

with

$$\begin{aligned} \mathcal{F}_{\text{qFM}}^{\text{hyb}} &= - \sum_{\alpha, X' \in \{F, A\}} 2 \text{Re}[g_{FX'}^{(\alpha)} \mathcal{X}_{X'}^{(\alpha)}], \\ \mathcal{F}_{\text{qFM}}^{\text{m-m}} &= - \sum_{X_1 X_2} K_{FX_1 X_2}^{\text{Fe}(3)} \mathcal{Q}_{X_1} \mathcal{Q}_{X_2} - \sum_{\alpha, X_1 X_2} K_{FX_1 X_2; \alpha}^{\text{FeTm}(3)} \mathcal{Q}_{X_1} \mathcal{X}_{X_2}^{(\alpha)}, \end{aligned} \quad (74)$$

and analogously for the qAFM and Tm flavor coordinates. The driving $\mathcal{F}^{\text{drv}}$ collects the linear-in-field THz couplings (in the geometries of [12] Sec. IX): for $H \parallel a$

it drives $\mathcal{Q}_{\text{qFM}}$; for $H \parallel c$ it drives $\mathcal{Q}_{\text{qAFM}}$; for $E \parallel c$ on Tm it drives $\mathcal{X}_F^{(\alpha)}$ (uniform CEF mode). Equation (73) is the desired Newtonian torque expression in canonical coordinates, valid to second order in the bosonic operators and exhibiting the four physical force channels (harmonic restoring, linear hybridization, magnon-magnon and magnon-flavor nonlinear, and external drive).

F. Recovery of $\chi^{(3)}$ 2DCS pathways

Equations (71)–(72) together with (73) are the bosonic analog of the joint Bloch-magnon nonlinear EOM of [12] Sec. XI–XII. Iterating perturbatively in $\mathcal{F}^{\text{drv}}$ to third order, with the cubic-vertex couplings of (68), (69), and (70) providing the wave-mixing, reproduces every double-sided Feynman pathway of the 2DCS peak catalog: each rephasing/non-rephasing peak at (ω_τ, ω_t) corresponds to one specific cubic-vertex contraction of two qFM/qAFM legs with one Tm flavor leg (or vice versa), with the Bertaut Klein-four selection rule controlling which (X_1, X_2, X_3) combinations are allowed in each geometry.

X. SUMMARY

We have presented the full bosonization scheme that the `ClassicalSpin_Cpp/TmFeO3` implementation realizes at the *classical-trajectory* level:

1. Fe $S = 5/2$ spins $\rightarrow$ HP magnons $a_{i,\mu}$ in local Bertaut frames, with the Klein-four sign matrix η_μ (2) absorbed by the rotation D_μ so the Γ_2 vacuum has uniform local- $+\hat{z}'$ polarization;
2. Tm $^{3+}$ qutrits $\rightarrow$ generalized SU(3) flavor-wave bosons $\{b_{i,\mu}^{(2)}, b_{i,\mu}^{(3)}\}$ with $b^{(1)}$ condensed on $|E_1\rangle$ via the multiboson HP [2–4];
3. χ and W Fe–Tm couplings $\rightarrow$ linear-in- b driving of Tm flavor bosons (only the $A_1^+ = \{1, 3, 8\}$ channels survive inversion at $\mathbf{k} = 0$; the $A_2^+ = \{4, 6\}$ channels are the genuinely staggered hybridization channels), plus bilinear hybridization $\sqrt{SM}(a + a^\dagger)(b + b^\dagger)$;
4. the resulting EOM is the paraunitary BdG equation (39) with the explicit 24×24 kernel (28).

The explicit hybridization vertices (37)–(38) together with the Bertaut sign structure (9) encoded in $\sigma_{\mu\nu}^a$ provide the microscopic dictionary needed to take the implementation’s mixed SU(2) $\oplus$ SU(3) dynamics, expand to linear order around the Γ_2 vacuum, and recover the experimentally observed magnon and Tm CEF bands measured in TmFeO₃ (see e.g. [9–11]).

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- [1] T. Holstein and H. Primakoff, *Field dependence of the intrinsic domain magnetization of a ferromagnet*, Phys. Rev. **58**, 1098 (1940).
 - [2] N. Papanicolaou, *Unusual phases in quantum spin-1 systems*, Nucl. Phys. B **305**, 367 (1988).
 - [3] A. Joshi, M. Ma, F. Mila, D. N. Shi, and F. C. Zhang, *Elementary excitations in magnetically ordered systems with orbital degeneracy*, Phys. Rev. B **60**, 6584 (1999).
 - [4] R. A. Muniz, Y. Kato, and C. D. Batista, *Generalized spin-wave theory: Application to the bilinear-biquadratic model*, Ann. Phys. (NY) **351**, 567 (2014).
 - [5] K. Penc and A. M. Läuchli, in *Introduction to Frustrated Magnetism*, edited by C. Lacroix, P. Mendels, and F. Mila (Springer, Berlin, 2011), Chap. 13 (SU(N) flavor-wave theory).
 - [6] J. H. P. Colpa, *Diagonalization of the quadratic boson Hamiltonian*, Physica A **93**, 327 (1978).
 - [7] G. F. Herrmann, *Magnetic resonances and susceptibility in orthoferrites*, Phys. Rev. **133**, A1334 (1964); J. Phys. Chem. Solids **24**, 597 (1963).
 - [8] R. L. White, *Review of recent work on the magnetic and spectroscopic properties of the rare-earth orthoferrites*, J. Appl. Phys. **40**, 1061 (1969); see also R. M. White, Rev. Mod. Phys. **37**, 1 (1965).
 - [9] R. V. Mikhaylovskiy, T. J. Huisman, A. I. Popov, A. K. Zvezdin, Th. Rasing, R. V. Pisarev, and A. V. Kimel, *Terahertz magnetization dynamics induced by femtosecond resonant pumping of Tm $^{3+}$ transitions in the multisublattice magnet TmFeO₃*, Phys. Rev. B **92**, 094437 (2015).
 - [10] T. N. Stanislavchuk *et al.*, *Far-infrared optical and magnetic resonance studies of TmFeO₃*, Phys. Rev. B **93**, 094403 (2016).
 - [11] K. Yamaguchi *et al.*, *Terahertz time-domain observation of spin reorientation in orthoferrite ErFeO₃*, Phys. Rev. Lett. **105**, 237201 (2010).
 - [12] Companion document, `docs/tmfeo3.notes.tex`, in the `ClassicalSpin_Cpp` repository.